Linear Algebra

M 408L

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1. Vector Spaces

1.1. Definition

Notation 1

$\mathbb{F}$ stands for either $\mathbb{R}$ or $\mathbb{C}$.

Definition 1

A group is a set with a binary operation $+$ such that for all $a, b, c \in G$:
1. associativity: $(a + b) + c = a + (b + c)$
2. commutativity: $a + b = b + a$
3. identity: $\exists 0 \in G \ \forall a \in G : 0 + a = a$
4. inverse: $\forall a \in G \ \exists b \in G : a + b = 0$

Notation 2

Now that the general of G and $+$ are understood it is given.

Definition 2

A group $+$ on F with no binary operation $\cdot$ and $+$ such that
1. distributive: $a \cdot (b + c) = a \cdot b + a \cdot c$
2. $\mathbb{F}$ $\cdot$ $\mathbb{F}$ $\cdot$ $\mathbb{F}$ commutative: $(a \cdot b) \cdot c = a \cdot (b \cdot c)$
3. identity: $\exists 1 \in F \ \forall a \in F : 1 \cdot a = a$
4. $\forall a \in F \setminus \{0\} \ \exists b \in F : a \cdot b = 1$

Notation 3

V is a vector space over $\mathbb{F}$ if
1. V is an abelian group
2. $\forall a \in \mathbb{F} \ \forall v \in V \ \exists ! \ a \cdot v \in V$
3. $\forall a, b \in \mathbb{F} \ \forall v \in V : (a \cdot b) \cdot v = a \cdot (b \cdot v)$
4. $\forall a, b \in \mathbb{F} \ \forall v, w \in V : a \cdot (v + w) = a \cdot v + a \cdot w$
5. $\forall a, b \in \mathbb{F} \ \forall v \in V : (a + b) \cdot v = a \cdot v + b \cdot v$
6. $\forall v \in V : 1 \cdot v = v$

Notation 4

$\forall v \in V : 0 \cdot v = 0$ and $\forall a \in \mathbb{F} : a \cdot 0 = 0$

1.2. Subspaces

Definition 3

A subspace of V is a subset U of V such that
1. $0 \in U$
2. $\forall u, v \in U : u + v \in U$
3. $\forall a \in \mathbb{F} \ \forall u \in U : a \cdot u \in U$

Notation 5

$U \subseteq V$ means U is a subset of V that is a subspace.

Notation 6

$U \subseteq V$ and $W \subseteq V$ implies $U + W \subseteq V$.

Notation 7

$U \subseteq V$ and $W \subseteq V$ implies $U \cap W \subseteq V$.

Notation 8

$U \subseteq V$ and $W \subseteq V$ implies $U \cup W \subseteq V$ if and only if $U \subseteq W$ or $W \subseteq U$.

Notation 9

$U \subseteq V$ and $W \subseteq V$ implies $U \cup W \subseteq V$ if and only if $U + W = U \cup W$.

Notation 10

$U \subseteq V$ and $W \subseteq V$ implies $U \cup W \subseteq V$ if and only if $U \cap W = \{0\}$.

Notation 11

$U \subseteq V$ and $W \subseteq V$ implies $U \cup W \subseteq V$ if and only if $U \cap W = U \cup W$.

Notation 12

$U \subseteq V$ and $W \subseteq V$ implies $U \cup W \subseteq V$ if and only if $U \cap W = \{0\}$ and $U + W = U \cup W$.

1.3. Span

Notation 13

Span is a map $\text{span} : \mathbb{F}^n \to \mathbb{F}^n$.

Notation 14

$\text{span}(v_1, \ldots, v_n) = \{a_1 v_1 + \cdots + a_n v_n : a_1, \ldots, a_n \in \mathbb{F}\}$.

Notation 15

$\text{span}(v_1, \ldots, v_n) \subseteq V$ is a subspace of V.

Notation 16

$\text{span}(v_1, \ldots, v_n)$ is the smallest subspace of V that contains $v_1, \ldots, v_n$.

Notation 17

$\text{span}(v_1, \ldots, v_n) = \{v_1, \ldots, v_n\}$ if and only if $v_1, \ldots, v_n$ is a subspace of V.

Notation 18

$\text{span}(v_1, \ldots, v_n) = V$ is a spanning set of V.

Notation 19

V is finite dimensional if $\exists v_1, \ldots, v_n \in V : \text{span}(v_1, \ldots, v_n) = V$.

Notation 20

V is infinite dimensional if $\forall v_1, \ldots, v_n \in V : \text{span}(v_1, \ldots, v_n) \neq V$.

1.4. Basis

Notation 21

A list $v_1, \ldots, v_n$ of vectors in V is a basis of V if
1. $\text{span}(v_1, \ldots, v_n) = V$
2. $\forall a_1, \ldots, a_n \in \mathbb{F} : a_1 v_1 + \cdots + a_n v_n = 0 \Rightarrow a_1 = \cdots = a_n = 0$

Notation 22

V is finite dimensional if V has a basis.

Notation 23

V is infinite dimensional if V has no basis.

Notation 24

V is finite dimensional if V has a spanning set.

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V is finite dimensional if V has a finite basis.

Notation 363

V is infinite dimensional if V has no finite basis.

Notation 364

V is finite dimensional if V has a finite spanning set.

Notation 365

V is infinite dimensional if V has no finite spanning set.

Notation 366

V is finite dimensional if V has a finite basis.

Notation 367

V is infinite dimensional if V has no finite basis.

Notation 368

V is finite dimensional if V has a finite spanning set.

Notation 369

V is infinite dimensional if V has no finite spanning set.

Notation 370

V is finite dimensional if V has a finite basis.

Notation 371

V is infinite dimensional if V has no finite basis.

Notation 372

V is finite dimensional if V has a finite spanning set.

Notation 373

V is infinite dimensional if V has no finite spanning set.

Notation 374

V is finite dimensional if V has a finite basis.

Notation 375

V is infinite dimensional if V has no finite basis.

Notation 376

V is finite dimensional if V has a finite spanning set.

Notation 377

V is infinite dimensional if V has no finite spanning set.

Notation 378

V is finite dimensional if V has a finite basis.

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V is infinite dimensional if V has no finite basis.

Notation 380

V is finite dimensional if V has a finite spanning set.

Notation 381

V is infinite dimensional if V has no finite spanning set.

Notation 382

V is finite dimensional if V has a finite basis.

Notation 383

V is infinite dimensional if V has no finite basis.

Notation 384

V is finite dimensional if V has a finite spanning set.

Notation 385

V is infinite dimensional if V has no finite spanning set.

Notation 386

V is finite dimensional if V has a finite basis.

Notation 387

V is infinite dimensional if V has no finite basis.

Notation 388

V is finite dimensional if V has a finite spanning set.

Notation 389

V is infinite dimensional if V has no finite spanning set.

Notation 390

V is finite dimensional if V has a finite basis.

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V is infinite dimensional if V has no finite basis.

Notation 392

V is finite dimensional if V has a finite spanning set.

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V is infinite dimensional if V has no finite spanning set.

Notation 394

V is finite dimensional if V has a finite basis.

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V is infinite dimensional if V has no finite basis.

Notation 396

V is finite dimensional if V has a finite spanning set.

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V is infinite dimensional if V has no finite spanning set.

Notation 398

V is finite dimensional if V has a finite basis.

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V is infinite dimensional if V has no finite basis.

Notation 400

V is finite dimensional if V has a finite spanning set.

Notation 401

V is infinite dimensional if V has no finite spanning set.

Notation 402

V is finite dimensional if V has a finite basis.

Notation 403

V is infinite dimensional if V has no finite basis.

Notation 404

V is finite dimensional if V has a finite spanning set.

Notation 405

V is infinite dimensional if V has no finite spanning set.

Notation 406

V is finite dimensional if V has a finite basis.

Notation 407

V is infinite dimensional if V has no finite basis.

Notation 408

V is finite dimensional if V has a finite spanning set.

Notation 409

V is infinite dimensional if V has no finite spanning set.

Notation 410

V is finite dimensional if V has a finite basis.

Notation 411

V is infinite dimensional if V has no finite basis.

Notation 412

V is finite dimensional if V has a finite spanning set.

Notation 413

V is infinite dimensional if V has no finite spanning set.

Notation 414

V is finite dimensional if V has a finite basis.

Notation 415

V is infinite dimensional if V has no finite basis.

Notation 416

V is finite dimensional if V has a finite spanning set.

Notation 417

V is infinite dimensional if V has no finite spanning set.

Notation 418

V is finite dimensional if V has a finite basis.

Notation 419

V is infinite dimensional if V has no finite basis.

Notation 420

V is finite dimensional if V has a finite span